

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

EE/EX-3005-CBGS

B.E., III Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Signals and Systems

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Sketch neat diagram.

1. a) Write short notes on followings:

i) Periodic and non-periodic signals

ii) Power and energy signals

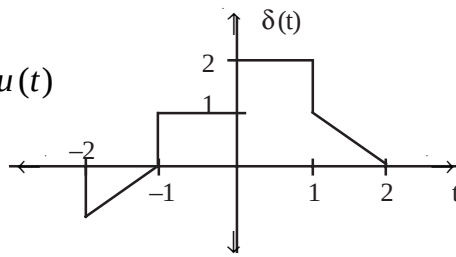
iii) Discrete time impulse function $\delta(n)$

b) A continuous time signal given below. Sketch the following signals.

i) $\delta\left[4 - \left(\frac{t}{2}\right)\right]$

ii) $[\delta(t) + \delta(-t)]u(t)$

iii) $\delta(2t + 1)$



2. a) Discuss the determination of the Fourier series representation of continuous time periodic signals.

b) Determine the fundamental frequency ω_0 and continuous-time Fourier series coefficient A_K for the signal.

$$\delta(t) = 2 + \cos\left(\frac{2\pi}{3}t\right) + 4\sin\left(\frac{5\pi}{3}t\right)$$

EE/EX-3005-CBGS

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[2]

3. a) List and explain various properties of Fourier transform.
b) What do you understand by region of convergence (ROC)? Explain with example.
4. a) Find Fourier transform of
 - i) $t \cos at$
 - ii) $\operatorname{sgn}(t)$b) Calculate the Laplace transform of the following.
 - i) $t \sin at$
 - ii) te^{-2t}
 - iii) t^n
5. a) State and prove time shifting and differentiation in the Z-domain property of Z-transform.
b) Determine the causal signal $x(n)$ whose Z-transform is given by
 - i) $x(z) = \frac{1+z^{-1}}{1-z^{-1}+0.5z^{-2}}$
 - ii) $x(z) = \frac{1}{(1+z^{-1})(1-z^{-1})^2}$
6. a) Determine the response of the system
$$y(n) = \frac{5}{6}y(n-1) - \frac{1}{6}y(n-2) + x(n)$$
to the input signal $x(n) = \delta(n) - \frac{1}{3}\delta(n-1)$
b) State and prove convolution property of Z-transform.

[3]

7. a) If the unit step response of a system starting from rest is given by $s(t) = 3(1 - e^{-st}), t \geq 0$. Then find the transfer function of the system.
- b) If $x(n) \{1, 5, -2, 1, \underset{\uparrow}{3}, 4, 2, 0, 5\}$ then find DTFT $\left[x(e^{j\omega}) \right]$.
8. Write short notes on the following.
- i) Classification of continuous and discrete time signals
 - ii) Linear time invariant systems and properties
 - iii) Fourier series analysis
